

Figure 1: Weka's features

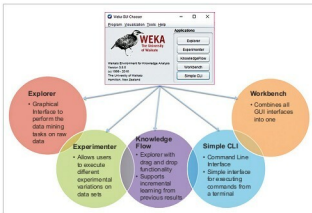


Figure 2: Weka's application interfaces

for Weka. When we open Weka, it will start the *Weka GUI Chooser* screen from where we can open the Weka application interface.

The Weka GUI screen and the available application interfaces are seen in Figure 2.

Weka data formats

Weka uses the Attribute Relation File Format for data analysis, by default. But listed below are some formats that Weka supports, from where data can be imported:

- CSV
- ARFF
- Database using ODBC

Attribute Relation File Format (ARFF): This has two parts:

- The header section defines the relation (data set) name, attribute name and the type.
- The data section lists the data instances.

An ARFF file requires the declaration of the relation, attribute and data. Figure 3 is an example of an ARFF file.

@relation: This is the first line in any ARFF file, written in the header section, followed by the relation/data set name. The relation name must be a string and if it contains spaces,

```
% Title: Database for fitting contact lenses
@relation lenses
@attribute age {young, pre-presbyopic, presbyopic}
@attribute spectprescrip {myope, hypermetrope}
@attribute astigmatism {no, yes}
@attribute tearprodrate {reduced, normal}
@attribute lenses {soft, hard, none}
@data
young,myope,no,reduced,none
young,myope,no,normal,soft
pre-presbyopic,myope,no,reduced,none
pre-presbyopic,myope,no,normal,soft
presbyopic,myope,no,normal,none
presbyopic,myope,yes,reduced,none
presbyopic,myope,yes,normal,hard
```

Figure 3: An example of an ARFF file

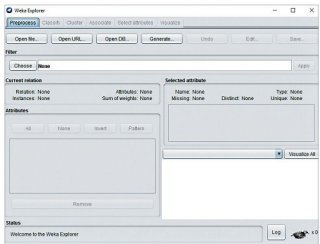


Figure 4: Weka Explorer

then it should be enclosed between quotes.

@attribute: These are declared with their names and the type or range in the header section. Weka supports the following data types for attributes:

- Numeric
- <nominal-specification>
- String
- date
- @data** – Defined in the *Data* section followed by the list of all data segments

Weka Explorer

The Weka Explorer is illustrated in Figure 4 and contains a total of six tabs.

The tabs are as follows.

- Preprocess:** This allows us to choose the data file.
- Classify:** This allows us to apply and experiment with different algorithms on preprocessed data files.
- Cluster:** This allows us to apply different clustering tools, which identify clusters within the data file.
- Association:** This allows us to apply association rules, which identify the association within the data.
- Select attributes:** These allow us to see the changes on the